test manual

1.1 System burning

Preparation before burning:

(1) Install Etcher software, download address https://www.balena.io/etcher.

(2) Download the ZYNQ image of Wildfire (3) A blank 2~32GB SD card. If there is content in the SD card, please back it up first (burning will clear the SD).

Open the installed Etcher tool. The software interface is as shown below:

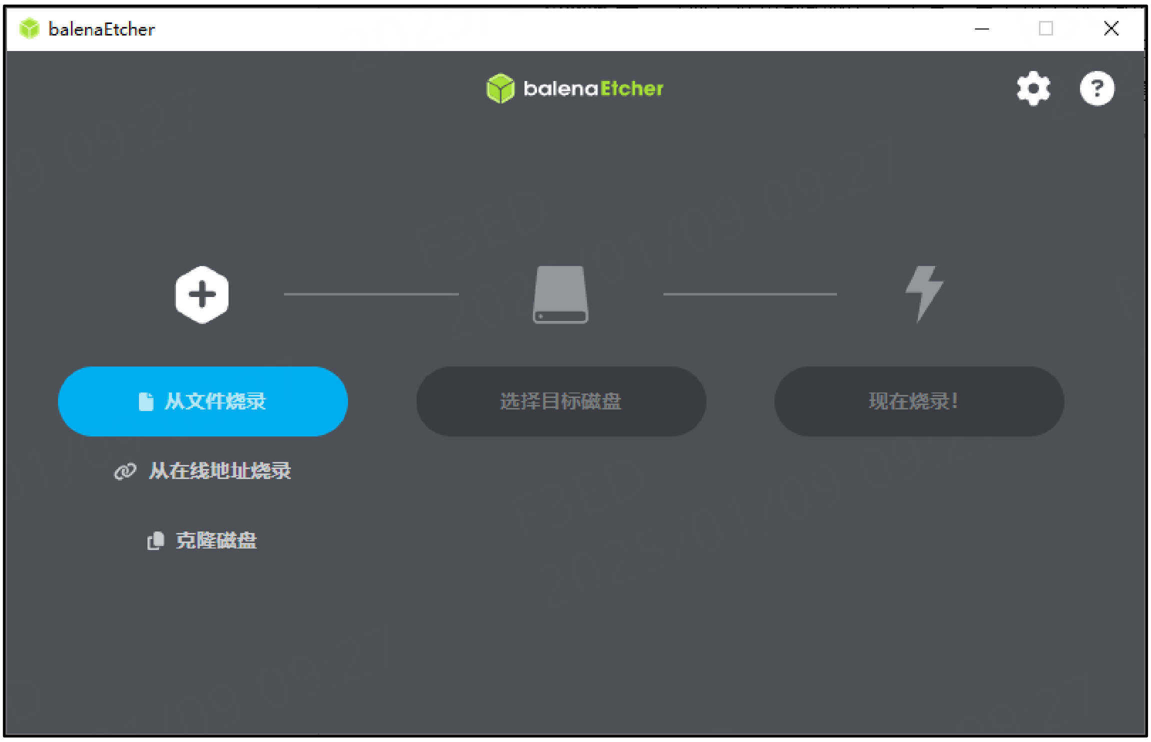


Figure 0-1 Etcher software interface

Click "Burn from file" and select the ZYNQ image provided by Yehuo. This image is adapted to the Haoyue development board and is common to both zynq7010 and zynq7020.

Figure 0-2 Wildfire Mirror

After selecting the zynq image, continue to select the SD card to be burned, and then click the "Burn Now" button to start burning.

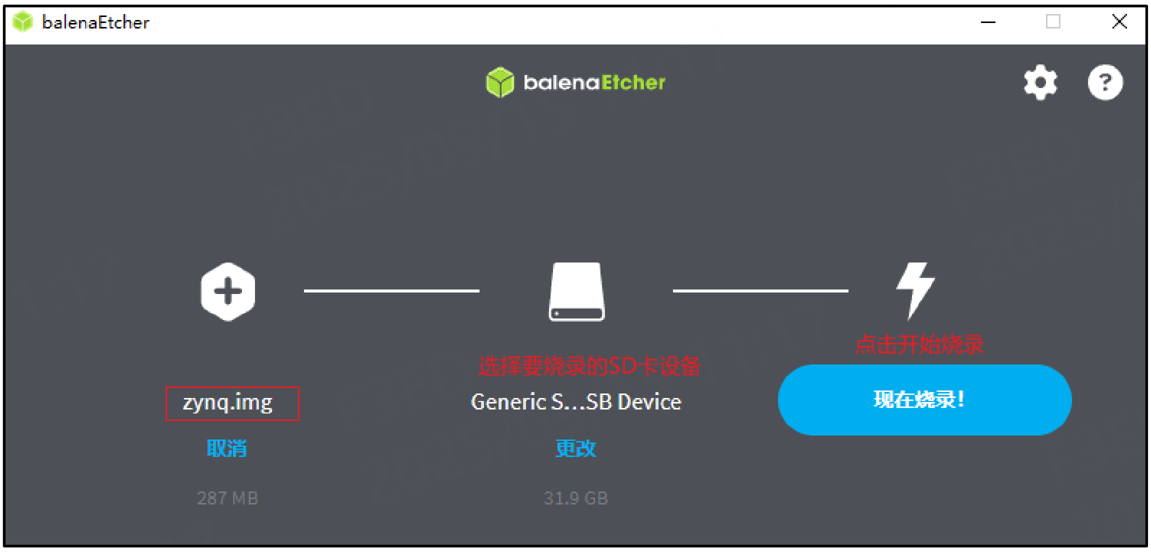


Figure 0-3 Start burning

When the following interface appears in the software, it means that the image has been burned successfully.

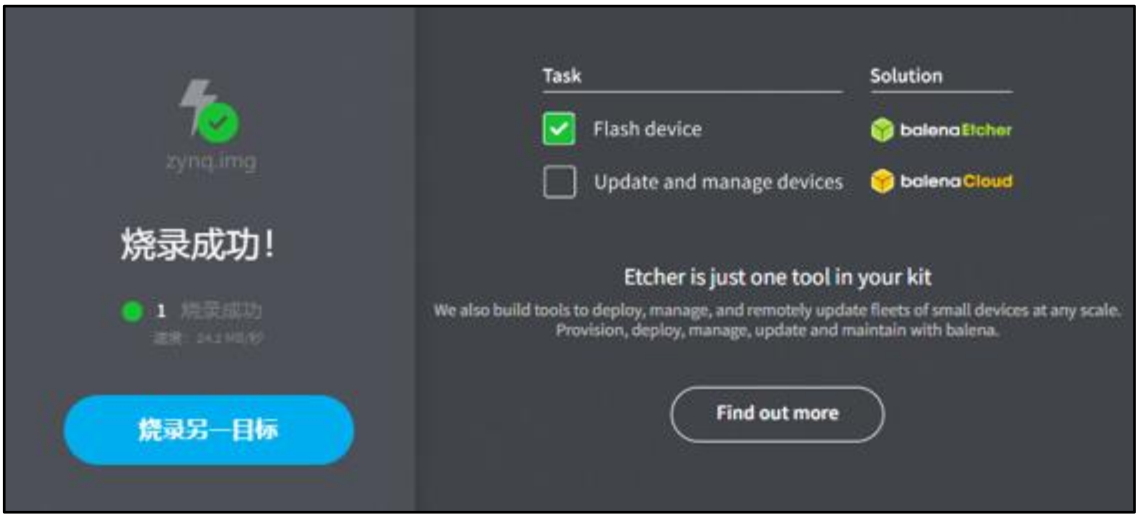


Figure 0-4 Burning completed

1.2 Run the development board and log in to the system

1.2.1 Serial terminal

As a terminal tool, we recommend the MobaXterm terminal software. It is very easy to use and po werful, and it also supports Chinese. The MobaXterm installation package has been placed in "3-Devel opment Software\MobaXterm_Installer_v12.3.zip" in the data disk.

Double-click to open the MobaXterm installation program and follow the prompts to complete the installation.

After completing the installation, use a USB cable to connect the USB-to-serial port of the com puter and development board, and then open the MobaXterm software. The software interface is as s hown below:

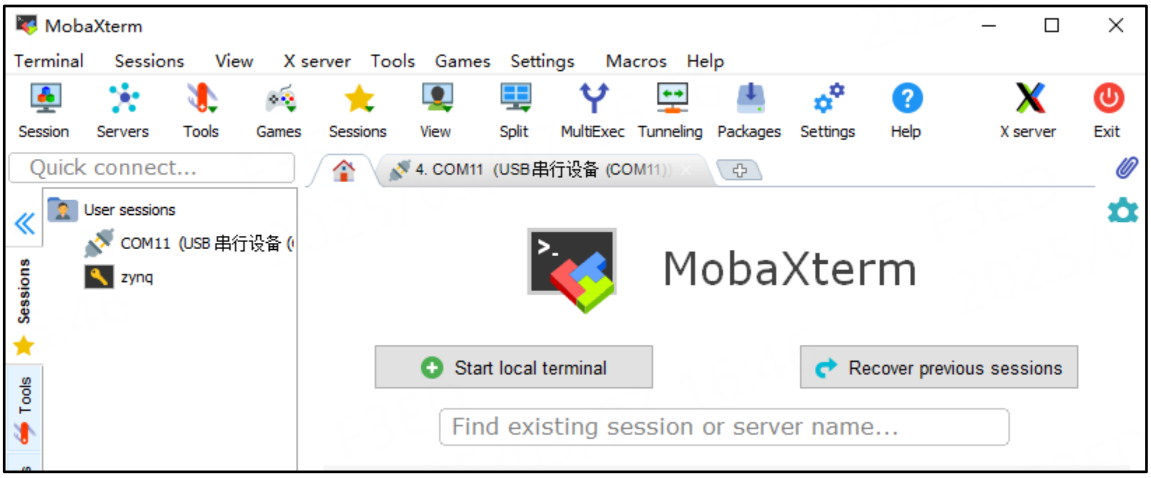


Figure 0-5 MobaXterm software interface

Note: When you install it for the first time, there is no session in the session tab bar on the left.

Click "sessions" -> "new session" in the menu bar to pop up the "session setting" dialog box. As you can see from the session dialog box, MobaXterm supports many connection methods. Here we use the serial port connection method.

Select the serial port we use, for example, the port number used in this example is "COM3". Note that the port number should be selected according to your own experimental environment, and then configure the communication rate of MobaXterm's serial port to the value used by the development board's serial port terminal. The default is 115200, as shown in the figure below:

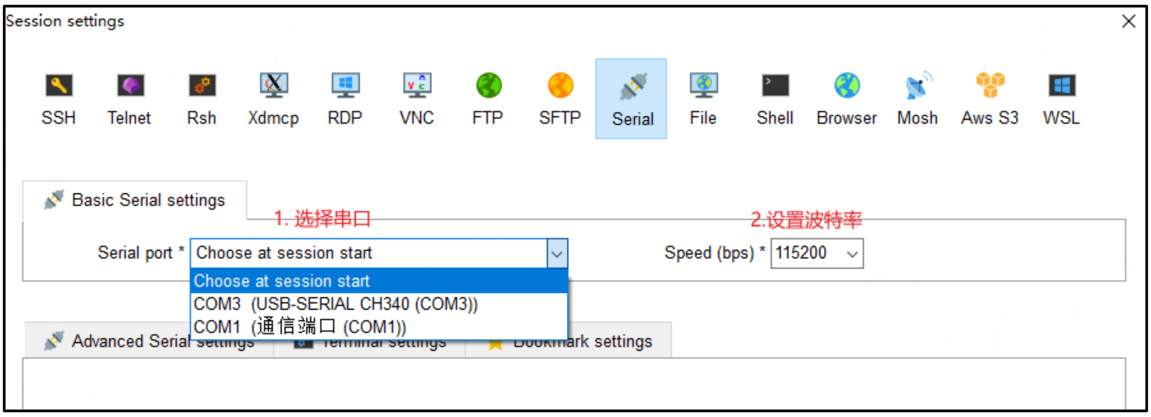


Figure 0-6 Create serial port connection

After selecting the serial port number and baud rate, click OK "" to complete the connection. The tab bar on the left will record this session, and you can open the session window directly from the tab bar in the future. As shown below:

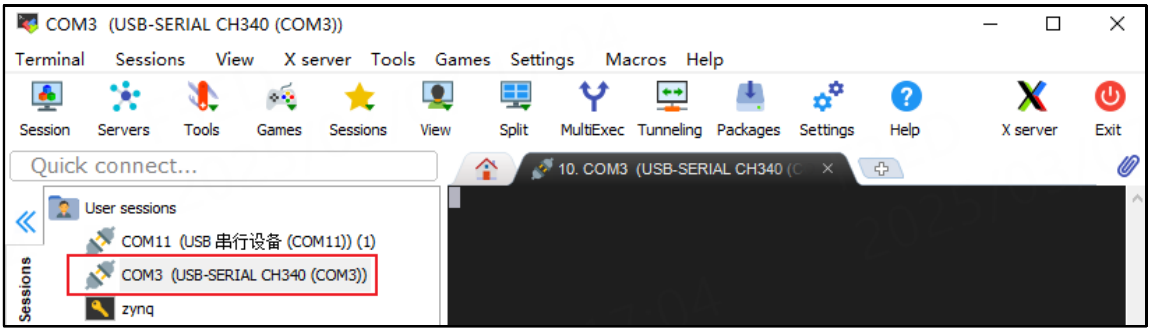


Figure 0-7 Session window

1.2.2 Development board operation

First, insert the SD card with the burned image into the development board, and adjust the DIP s witch to the SD card startup mode (MIO4 and MIO5 are both 1). Then use a USB cable to connect th e USB-to-serial port of the computer and development board. Finally, connect the development boar d to the 12V power supply and turn on the power switch.

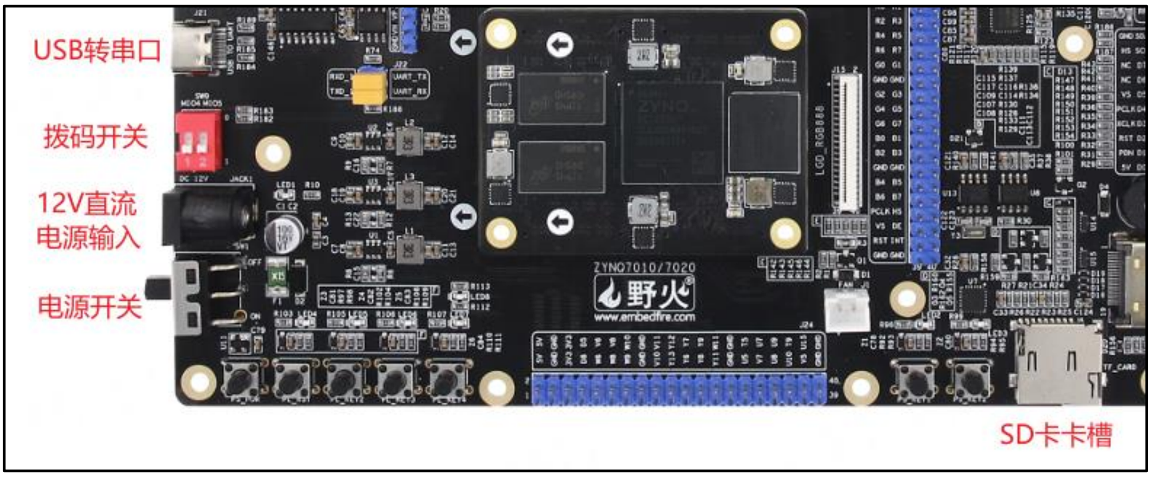


Figure 0-8 Start the development board

After the board power is started, you will see the information output of the development board when starting, see the picture below:

```
random: ln: uninitialized urandom read (6 bytes read)
Starting internet superserver: inetd.
NFS daemon support not enabled in kernel
Starting ntpd: done
Starting syslogd/klogd: done
Starting tcf-agent: random: crng init done
OK

PetaLinux 2022.2_release_S10071807 zynq_plnx /dev/ttyPS0

zynq_plnx login:
```

Figure 0-9 Development board startup information

Then enter your username and password to log in. There are two accounts in the system by default: Account name: root, password: root Account name: petalinux, pa ssword: ****** For petalinux users, you will be prompted to change the password w hen you log in for the first time, just follow the prompts. The following takes root use r login as an example, see the figure below:

```
PetaLinux 2022.2_release_S10071807 zynq_plnx /dev/ttyPS0

zynq_plnx login: root
Password:
root@zynq_plnx:~#ls /
bin   dev  home  log_lock.pid  media  opt   run   sys  usr
boot  etc  lib   lost+found    mnt    proc  sbin  tmp  var
root@zynq_plnx:~#
```

Figure 0-10 Entering the system

Note: The contents of this manual are all logged in using the root user by default.

1.3 Testing

1.3.1 Individual testing

Next, each peripheral of the Haoyue development board is tested individually (there is also an i ntroduction to the overall test below. If you want to directly conduct the overall test, you can directly see the overall test summary)

1. LED test

There are a total of 6 user LED lights on the development board, as follows:

Figure 0-11 LED on the development board

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/led_tester.sh
```

If all 6 user LEDs are flashing, the test is passed.

The terminal information is printed as follows:

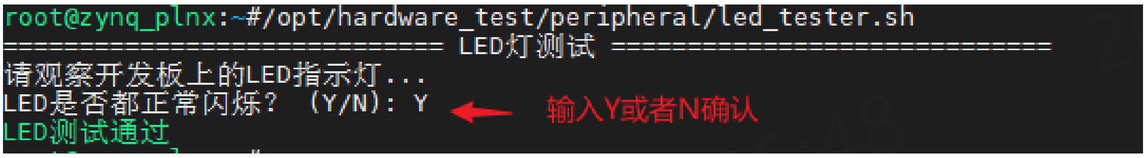


Figure 0-12 LED test terminal output

2. Button test

There are seven buttons in total on the development board, namely PS_POR, PL_RST (also call ed PL_KEY1), PL_KEY2, PL_KEY3, PL_KEY4, PS_KEY1, and PS_KEY2. Among them, PS_PO R is the system reset button, PL_KEY1~PL_KEY4 is the PL side button, and PS_KEY1~PS_KEY2 i s the PS side button.

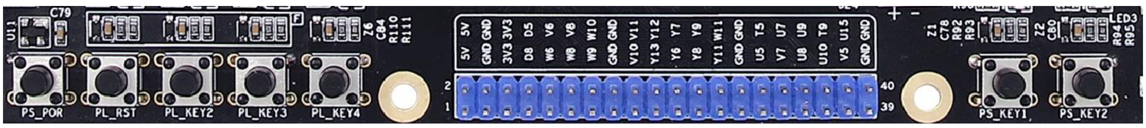


Figure 0-13 Buttons on the development board

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/key_tester.sh
```

During the test, you can press PL_KEY1~PL_KEY4, PS_KEY1~PS_KEY2 in sequence. When the keys are pressed or released, observe whether the terminal prints out the pressed or released infor mation, as shown in the following figure:

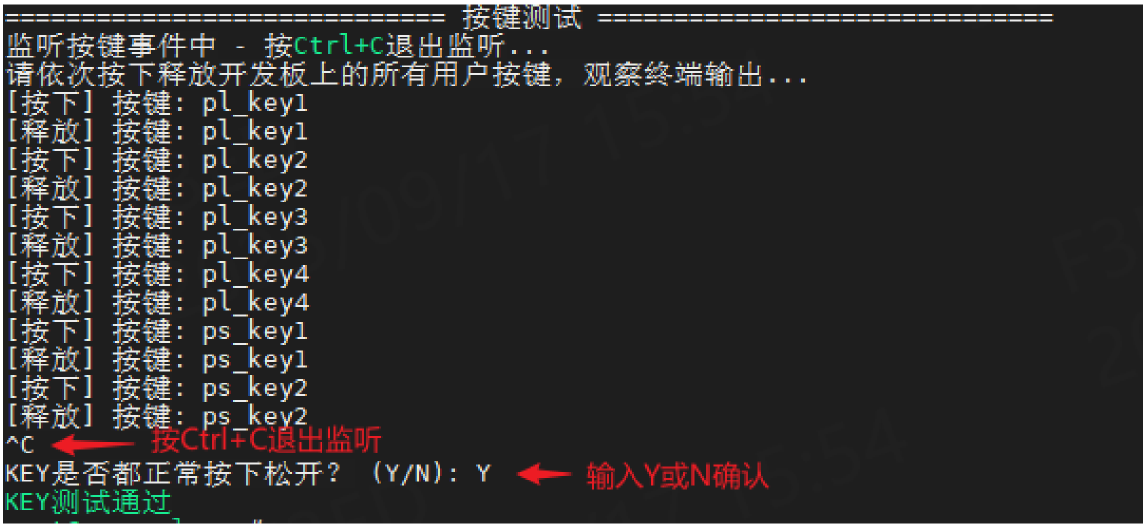


Figure 0-14 Button test

3. Buzzer test

The buzzer on the development board is as shown below.

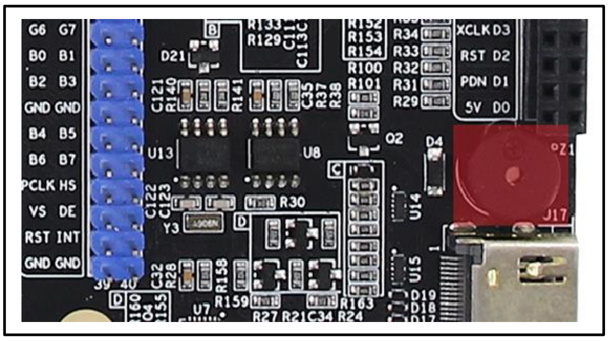


Figure 0-15 Buzzer

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/beep_tester.sh
```

After the program runs, the buzzer will sound three times.

4. Fan interface test

The fan interface on the base plate is as shown in the figure below:

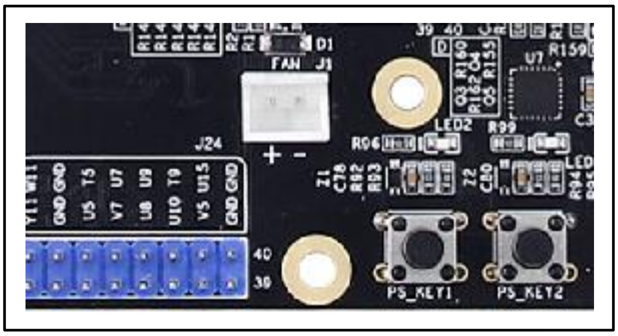


Figure 0-16 Fan interface

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/fan_tester.sh
```

The test will start at full speed, then switch speeds every 3 seconds, and finally the fan will stop spinning.

5. RTC test

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/rtc_tester.sh
```

The normal test phenomena are as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/rtc_tester.sh
============================== RTC测试 ==============================
设定时间: 2024-10-10 12:00:00
当前时间: 2024-10-10 12:00:01
RTC测试通过
```

Figure 0-17 RTC test

6. FLASH test

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/flash_tester.sh
```

The normal test phenomena are as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/flash_tester.sh
============================= flash测试 =============================
写flash (mtdblock1).........................
flash写入成功
读flash (mtdblock1)....
flash读取成功
数据验证...
flash测试通过
```

Figure 0-18 FLASH test

7. EEPROM test

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/eeprom_tester.sh
```

The normal test phenomena are as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/eeprom_tester.sh
============================= eeprom测试 ============================
写EEPROM.....
EEPROM写入成功
读EEPROM....
EEPROM读取成功
数据验证...
EEPROM测试通过
```

Figure 0-19 EEPROM test

8. EMMC test

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/eeprom_tester.sh
```

The test phenomenon is as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/emmc_tester.sh
============================== EMMC测试 =============================
未检测到分区，检查设备是否已格式化
设备未格式化，正在创建ext4文件系统...
挂载：成功
-------- 顺序写入测试 -------
eMMC写入速度: 19.3 MB/s
-------- 顺序读取测试 -------
eMMC读取速度: 24.0 MB/s
emmc读写测试通过
```

Figure 0-20 EMMC test

Normally, "emmc reading and writing test passed" will be printed.

9. SD card test

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/sd_tester.sh
```

The test phenomenon is as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/sd_tester.sh
============================== SD卡测试 =============================
检测到已插入SD卡
检测到分区表，使用最后一个分区进行测试
挂载：成功
-------- 顺序写入测试 -------
SD卡写入速度: 16.9 MB/s
-------- 顺序读取测试 -------
SD卡读取速度: 23.7 MB/s
sd卡读写测试通过
root@zynq_plnx:~#
```

Figure 0-21 SD card test

Normally, "sd card read and write test passed" will be printed.

10. XADC test

The VP/VN interface on the development board is as follows:

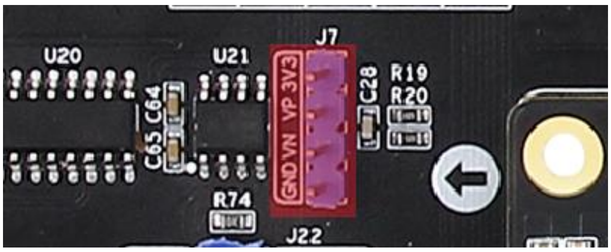


Figure 0-22 VP/VN interface

In the provided system image, VP/VN is configured as unipolar, so connect VN to the GND pin of the development board, and VP to the positive analog input voltage (voltage range is 0~1V). It is necessary to keep the negative electrode of the analog input voltage connected to the GND pin of the development board first to meet the common ground requirement.

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/xadc_tester.sh
```

测试现象如下：

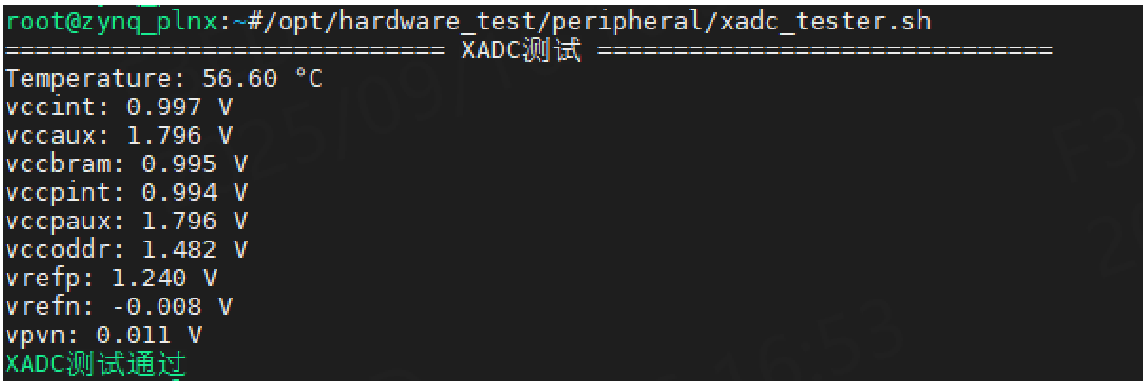


Figure 0-23 XADC test

11. USB test

There are four usb host interfaces and one usb slave interface on the zynq development board. Four of the hsot interfaces are expanded through the hub chip. And because the slave interface and the USB host interface cannot be used at the same time, they are in host mode by default. When the slave interface is connected to the host, it will automatically switch to slave mode.

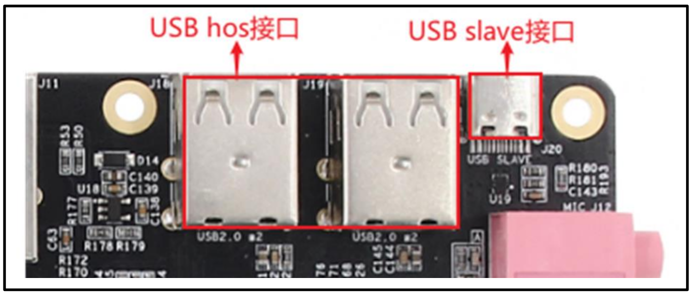


Figure 0-24 usb interface

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/usb_tester.sh
```

测试现象如下：

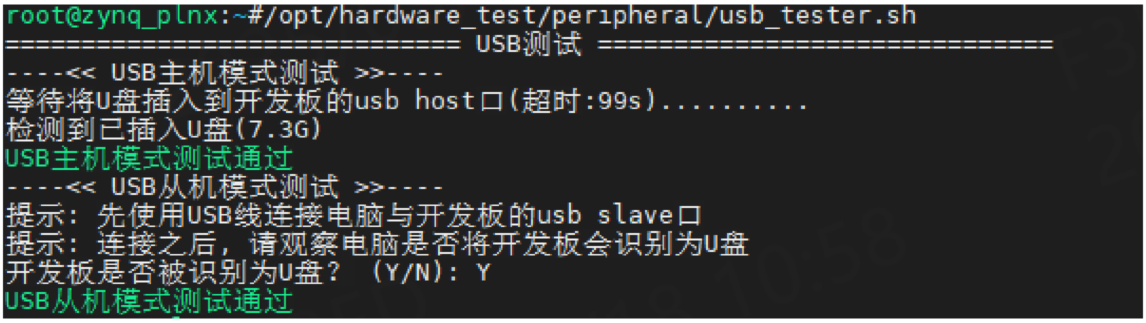


Figure 0-25 USB test

USB host mode test: Disconnect the USB SLAVE interface first, and then insert the U disk into the USB HOST interface. If the U disk and its size can be recognized, the USB HOST interface is no rmal.

Figure 0-26 Insert U disk

USB slave mode test: Use a USB cable to connect the computer to the USB SLAVE port of the development board. After the connection, the computer will recognize the development board as a U disk device under normal circumstances.

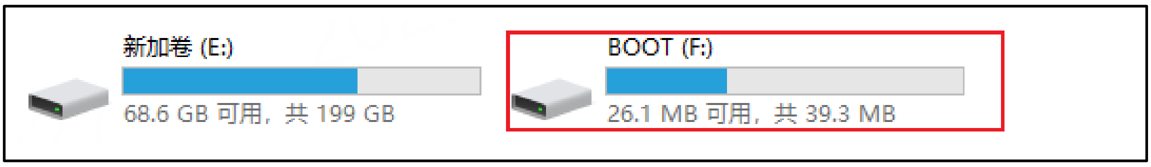


Figure 0-27 The development board is recognized as a USB flash drive

12. Network port test

The network port resources on the development board are as follows:

Figure 0-28 Gigabit network port

When the test starts, the program prompts to run the: iperf3 -s command on the iperf3 server. Th e iperf3 server can be our computer. Below we use the computer as an iperf3 server as an example to demonstrate:

First, you need to install the iperf3 tool on your computer windows system. The iperf3.9_64.zip provided under the development software folder in the information is the iperf3 tool. You can use it after unzipping it. Here is an example of decompressing to drive D, as shown below:

Figure 0-29 iperf3

Before using iperf3, the development board and the computer should be in the same LAN, that is, the computer and the development board can ping each other. Press the win + R shortcut key, then enter cmd to open the computer command prompt window, and enter the ipconfig command to view the computer IP address.

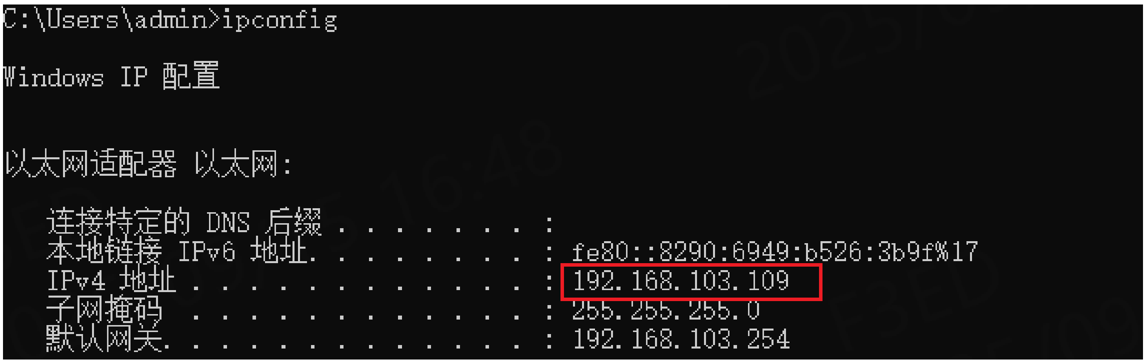


Figure 0-30 View computer IP

Run the iperf3 -s command on your computer to start the iperf3 service:

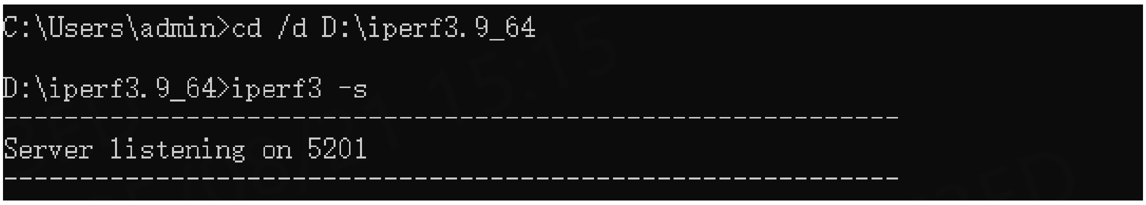


Figure 0-31 Computer as iperf3 server

Then run the following command on the development board to test:

```
/opt/hardware_test/peripheral/ethernet_tester.sh
```

The test program will prompt you to enter the IP address of the iperf3 server. We use the computer as the iperf3 server, so just enter the IP address of the computer, as shown below:

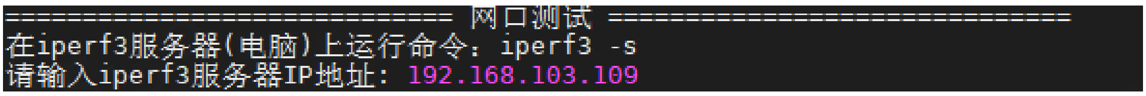


Figure 0-32 Enter computer IP

The test phenomenon is as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/ethernet_tester.sh
============================== 网口测试 ==============================
在iperf3服务器(电脑)上运行命令: iperf3 -s
请输入iperf3服务器IP地址: 192.168.103.109
----<< PS网口测试 >>----
PS网口的网线已连接
自动获取IP中.........
成功获取IP:192.168.103.155
---- PS网口发送测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PS网口发送速度: 807Mbits/sec
---- PS网口接收测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PS网口接收速度: 805Mbits/sec
PS网口测试通过
----<< PL网口测试 >>----
请把网线插到PL网口中
等待网线连接.................
PL网口的网线已连接
自动获取IP中..........
成功获取IP:192.168.103.152
---- PL网口发送测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PL网口发送速度: 807Mbits/sec
---- PL网口接收测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PL网口接收速度: 811Mbits/sec
PL网口测试通过
```

Figure 0-33 Network port test phenomenon

The program tests the PS network port first and then the PL network port. If it is detected that the network cable is not connected during the test, the program will prompt that the network cable needs to be inserted into the network port. Note: The other end of the network cable needs to be connected to the router, otherwise the IP cannot be obtained.

13. LCD display test

The screen used in the test is a 5-inch capacitive screen from Wildfire. Please note that the deve lopment board cannot be powered on when connected. It can only be powered on and started after co nfirming that the connection is complete. The connection diagram is as follows:

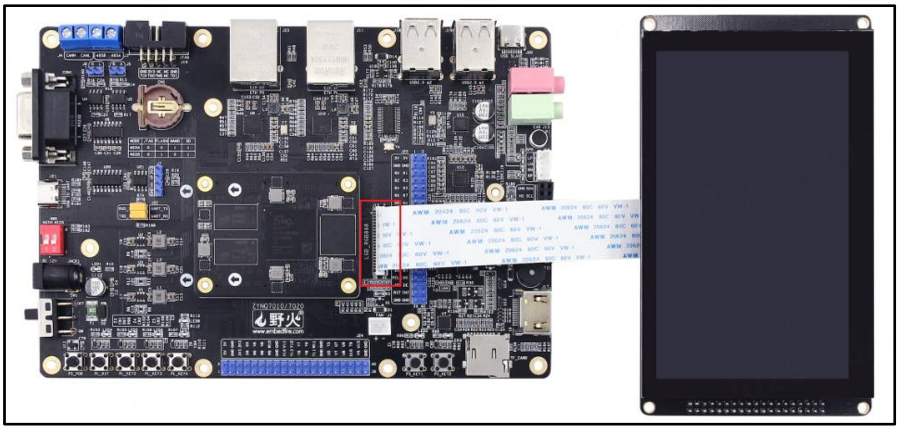


Figure 0-34 Connecting LCD screen

Since the system disables LCD display by default, you need to run the following command to e nable LCD display (resolution is 800*480):

```
/opt/hardware_test/config_display.sh lcd
```

This program will check the current system display. If the current system is not LCD display, it will modify the configuration file to enable LCD display and restart the system to take effect.

Then run the following command on the development board to test:

```
/opt/hardware_test/peripheral/lcd_tester.sh
```

The normal display effect of the LCD screen is as follows:

Figure 0-35 LCD screen display test

14. Touch test

Before powering on the system, you need to connect the Yefire 5-inch capacitive touch screen. For the connection method,

The touch test also requires the system to enable LCD display. You can run this command to check whether the curr ent system has LCD display:

```
/opt/hardware_test/config_display.sh lcd
```

This program will check the current system display. If the current system is not LCD display, it will modify the configuration file to enable LCD display and restart the system to take effect.

Then run the following command on the development board for touch testing:

```
/opt/hardware_test/peripheral/touch_tester.sh
```

At the beginning of the test, you need to click "Draw" on the screen to enter the drawing mode, and then press and slide with multiple fingers. The test effect is as follows:

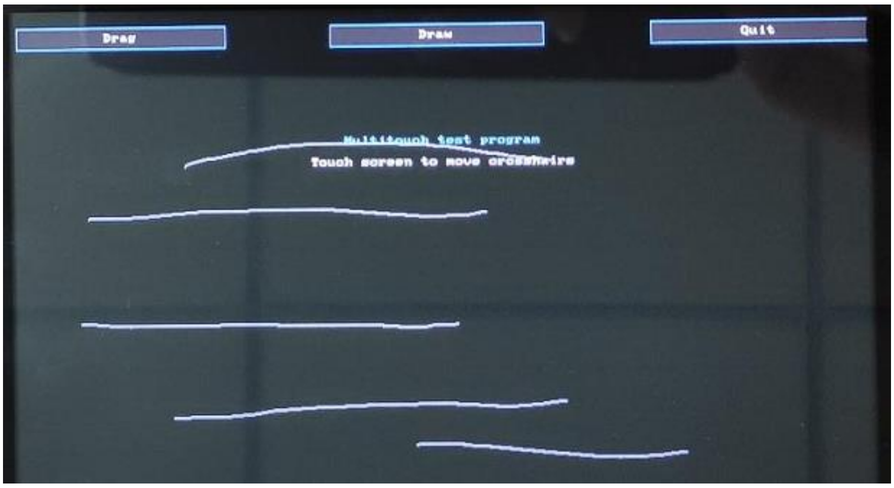


Figure 0-36 Touch test

After the test is completed, you need to click "Quit" on the screen to exit the touch screen test.

15. HDMI display test

Before testing, make sure the development board is connected to an HDMI screen.

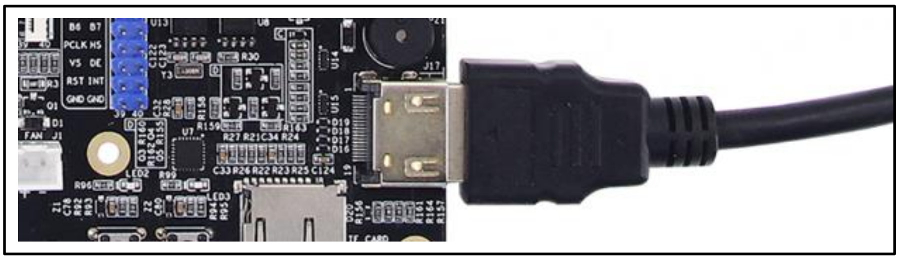


Figure 0-37 Connect HDMI screen

The system enables HDMI display by default. You can run this command to check whether the current system suppor

```
/opt/hardware_test/config_display.sh hdmi
```

If it is checked that the current system does not support HDMI display, the configuration file wi ll be modified to enable HDMI display and the system will be restarted to take effect.

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/hdmi_tester.sh
```

The normal display effect of HDMI is as follows:

Figure 0-38 HDMI display

16. Audio test

Instructions for developing audio interfaces:

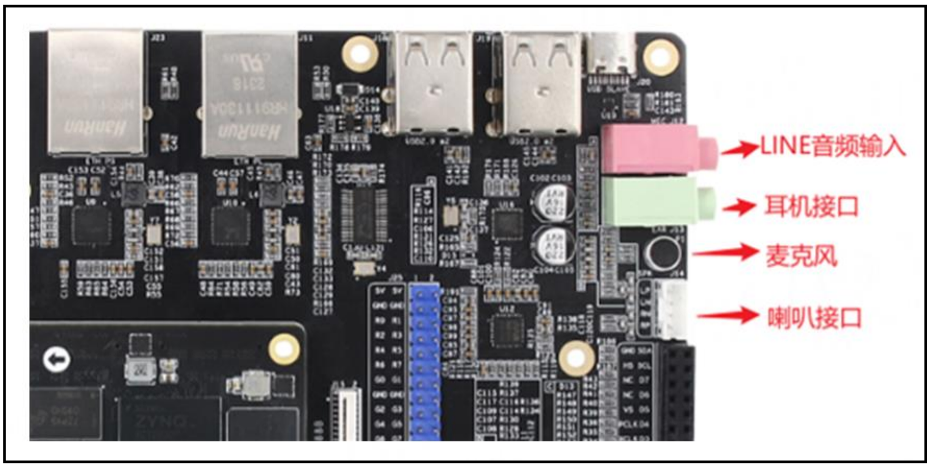


Figure 0-39 The actual audio interface of the development board

The development board is not equipped with speakers. When playing audio, you need to connect your own external speakers. Two 8Ω 1W speakers are supported.

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/hdmi_tester.sh
```

测试现象如下：

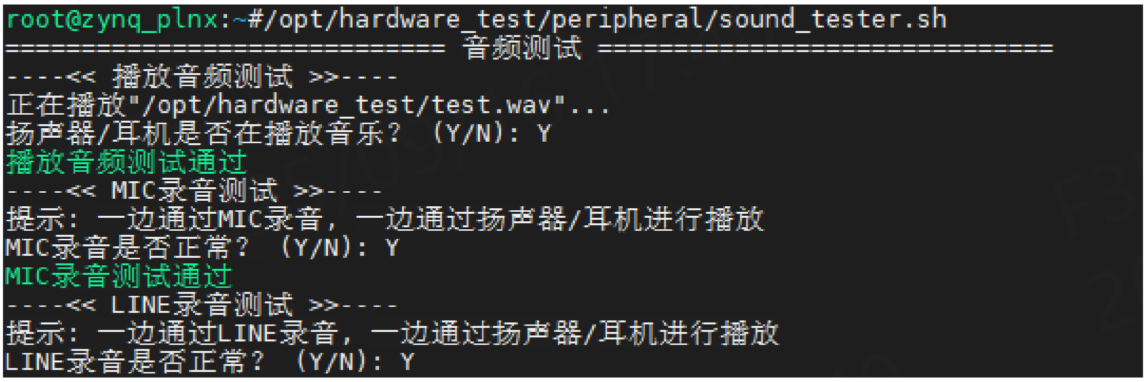


Figure 0-40 Audio test

Play audio test: If the sound card is normal, you can hear music playing from the speakers/head phones. The development board supports automatic switching between speakers and headphones. W hen no headphones are plugged in, the audio is output from the speakers. When headphones are plug ged in, the audio is output from the headphones instead.

MIC recording test: record and play at the same time.

LINE recording test: You need to use an audio cable to connect. For example, to output the com puter's audio to the development board, you need to use an audio cable to connect the development b oard to the computer, and then you can hear the audio played on the computer through the speakers/h eadphones.

17. Camera test

The camera used in the test program is ov5640. Since the camera does not support hot swapping, t he camera must be plugged into the development board before the system starts.

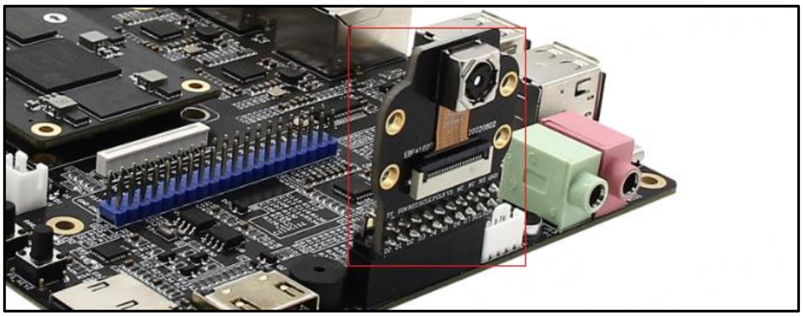


Figure 0-41 Connect camera

Run the following commands on the development board for testing:

```
/opt/hardware_test/peripheral/hdmi_tester.sh
```

测试的分辨率为 800*480，LCD 显示摄像头图像效果如下：

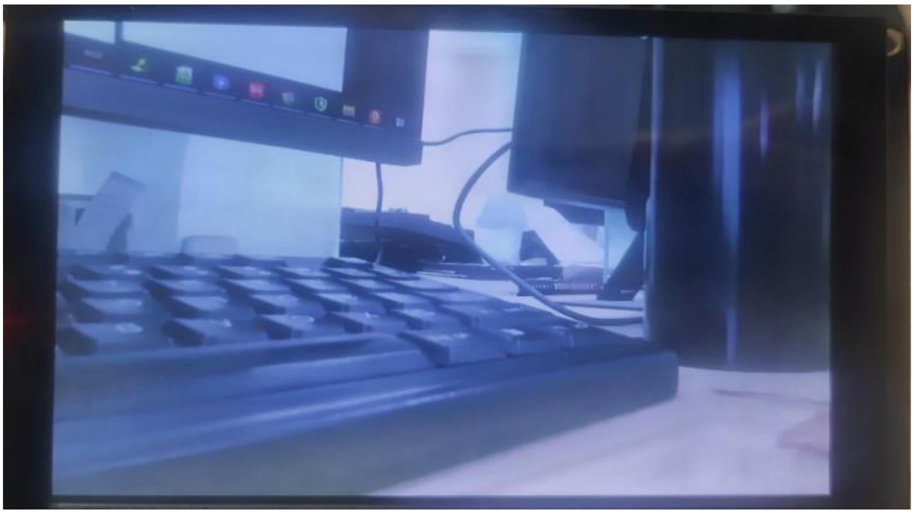


Figure 0-42 Camera real-time display

For zynq7010, when using the zynq7010-hdmi_tester.sh test program, the image captured by the camera will be displayed on the HDMI screen.

18. RS232 test

Before the experiment, use the RS232 cable to connect the development board to the computer, as shown in the figure

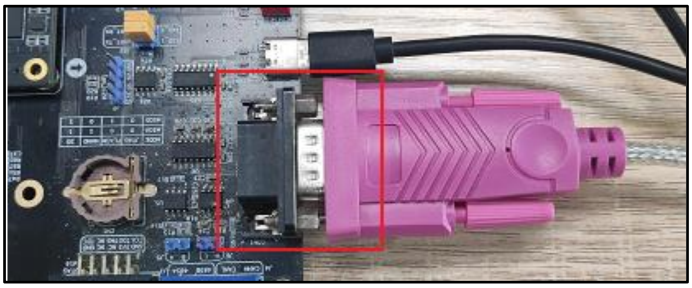


Figure 0-43 RS232 connection

First open the serial port assistant and connect to the serial port corresponding to RS232.

Figure 0-44 Connecting the serial port

Computers with different port numbers will have different port numbers. You need to choose according to the actual situation of your computer. The baud rate can be 115200.

Then the development board runs the following test program:

/opt/hardware_test/peripheral/rs232_tester.sh

At this time, the serial port assistant will receive the data sent by the development board, and then the serial port assis

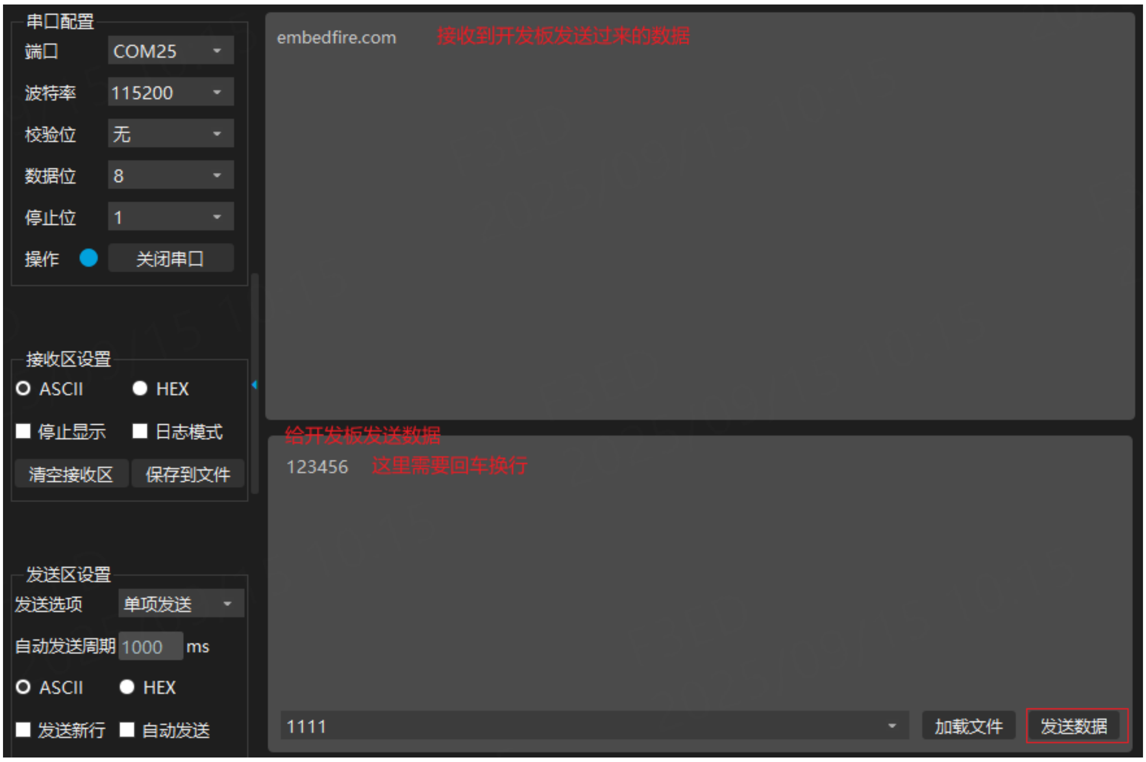


Figure 0-45 Serial port assistant sends and receives data

The developed sending and receiving data is as follows:

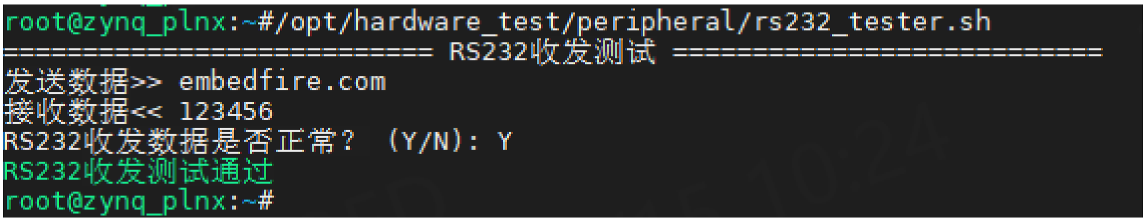


Figure 0-46 Development board RS232 sending and receiving data

19. RS485 test

Below we use two zynq development boards for RS485 communication testing. We need to con nect the RS485 terminals A of the two development boards to terminal A and terminal B to terminal B, as shown in the figure below:

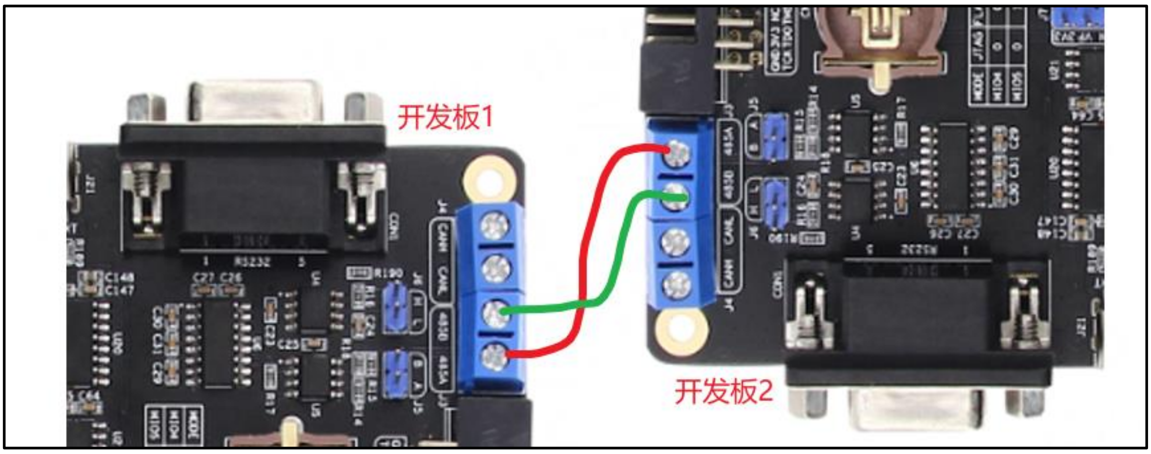


Figure 0-47 RS485 connection

First run the following command on development board 2: /opt/hardware_test/peripheral/rs485tester_board2.sh Then run the following command on board 1: /opt/hardware_test/peripheral/rs485tester_board1.sh

The test phenomenon of development board 1 is as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/rs485tester_board1.sh
=========================== RS485收发测试 ===========================
发送数据>> 1234567
接收数据<< abcdefg
RS485收发数据是否正常？ (Y/N): Y
RS485收发测试通过
root@zynq_plnx:~#
```

Figure 0-48 RS485 sending and receiving data of development board 1

The test phenomenon of development board 2 is as follows:

```
root@zynq_plnx:~#/opt/hardware_test/peripheral/rs485tester_board2.sh
=========================== RS485收发测试 ===========================
接收数据<< 1234567
发送数据>> abcdefg
RS485收发数据是否正常？ (Y/N): Y
RS485收发测试通过
root@zynq_plnx:~#
```

Figure 0-49 RS485 sending and receiving data on development board 2

20. CAN test

Use two ZYNQ development boards for CAN bus communication testing. The two ZYNQ development boards are connected to CANL and CANH, as shown in the figure below:

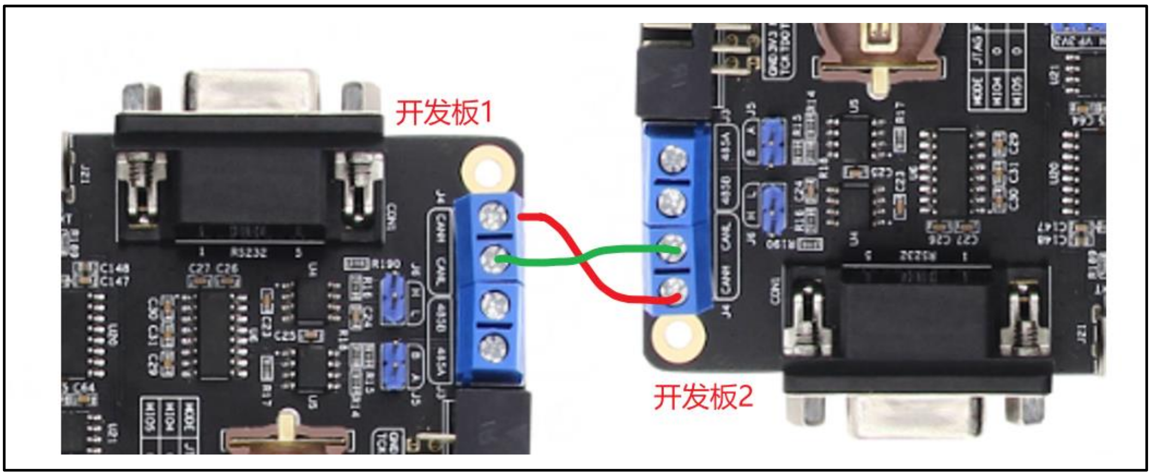


Figure 0-50 CAN bus connection

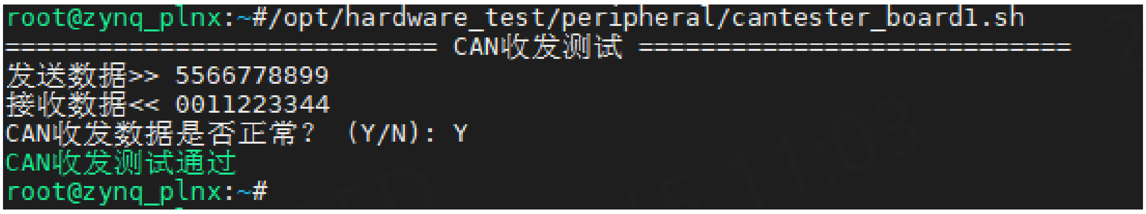
First run the following command on development board 2:

```
/opt/hardware_test/peripheral/cantester_board2.sh
```

Then run the following command on board 1:

```
/opt/hardware_test/peripheral/cantester_board1.sh
```

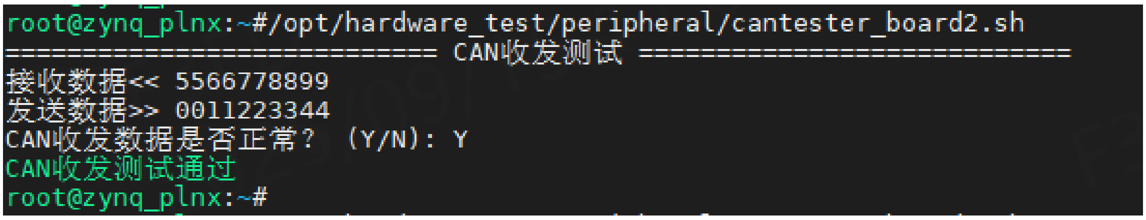
Test phenomenon of development board 1:



```
root@zynq_plnx:~#/opt/hardware_test/peripheral/cantester_board1.sh
============================ CAN收发测试 ============================
发送数据>> 5566778899
接收数据<< 0011223344
CAN收发数据是否正常？ (Y/N): Y
CAN收发测试通过
root@zynq_plnx:~#
```

Figure 0-51 CAN sending and receiving data of development board 1

Test phenomenon of development board 2:



```
root@zynq_plnx:~#/opt/hardware_test/peripheral/cantester_board2.sh
============================ CAN收发测试 ============================
接收数据<< 5566778899
发送数据>> 0011223344
CAN收发数据是否正常？ (Y/N): Y
CAN收发测试通过
root@zynq_plnx:~#
```

Figure 0-52 CAN sending and receiving data of development board 2

1.3.2 Overall test

In the development board system, we provide the overall test program: /opt/hardware_test/zynq_tester.sh. In fact, the overall test program just integrates each part of the individual test. If there is a problem with a certain peripheral during the test, you can refer to the corresponding content in the "Individual Test" section and test again. This way, you do not need to follow the overall test process and test again, which greatly saves testing time.

For zynq7020, it can support both LCD and HDMI displays. However, for Zynq7010, due to insufficient PL-side resources of Zynq7010, LCD display and HDMI display cannot be realized at the same time, and only one of them can be supported. By default, the system only enables HDMI display, and the LCD display is turned off, so you can run the following command to configure the system display before testing.

```
#For zyqn7020, run this command to enable LCD and HDMI display
/opt/hardware_test/config_display.sh lcd hdmi

#For zyqn7010, run this command to enable LCD display and turn off
HDMI display /opt/hardware_test/config_display.sh lcd

#For zyqn7010, run this command to enable HDMI and turn off LC
D display /opt/hardware_test/config_display.sh hdmi
```

Just run one of the commands. These commands will first check the current system display. If the current system display is not what we configured, the system display will be automatically modified and the system will be restarted to take effect. Take zynq7020 as an example, as shown below:

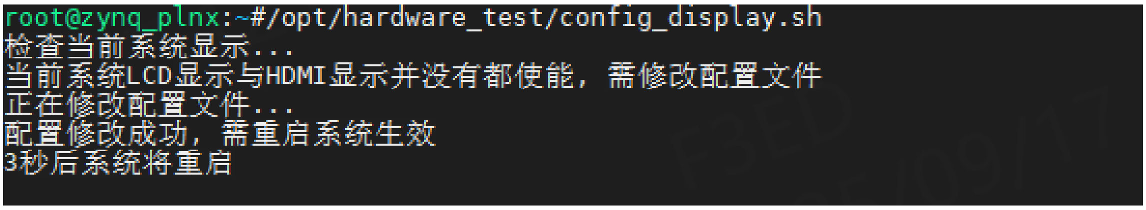


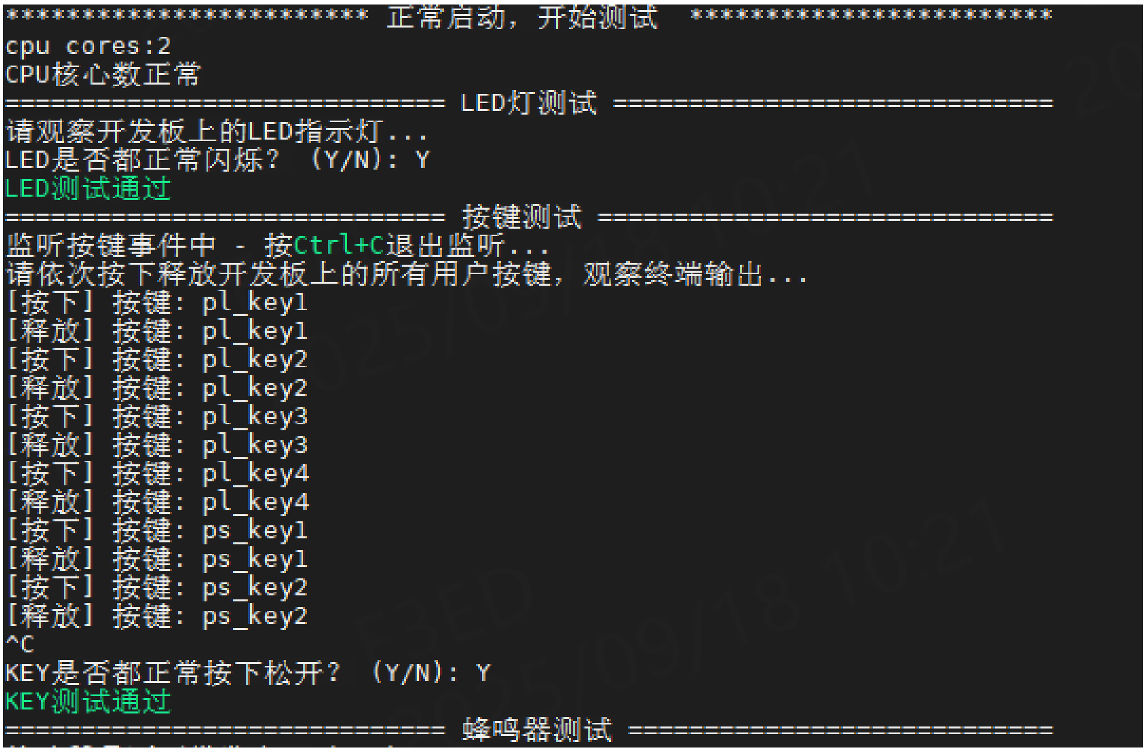
Figure 0-53 Configure system display

After the configuration is completed, the test can be started. Since it is an overall test, the HDMI, Wildfire 5-inch capacitive screen, ov5640 camera, speakers/headphones, etc. need to be connected before powering on the development board. Instructions for connecting these devices can be found in the "Individual Testing" section.

Run the following command on the development board:

```
/opt/hardware_test/zynq_tester.sh
```

进入测试之后根据提示完成即可，以 zynq7020 为例，其测试如下：



```
************************* 正常启动，开始测试 *************************
cpu cores:2
CPU核心数正常
============================== LED灯测试 ==============================
请观察开发板上的LED指示灯...
LED是否都正常闪烁？ (Y/N): Y
LED测试通过
============================== 按键测试 ==============================
监听按键事件中 - 按Ctrl+C退出监听...
请依次按下释放开发板上的所有用户按键，观察终端输出...
[按下] 按键: pl_key1
[释放] 按键: pl_key1
[按下] 按键: pl_key2
[释放] 按键: pl_key2
[按下] 按键: pl_key3
[释放] 按键: pl_key3
[按下] 按键: pl_key4
[释放] 按键: pl_key4
[按下] 按键: ps_key1
[释放] 按键: ps_key1
[按下] 按键: ps_key2
[释放] 按键: ps_key2
^C
KEY是否都正常按下松开？ (Y/N): Y
KEY测试通过
============================== 蜂鸣器测试 ============================
```

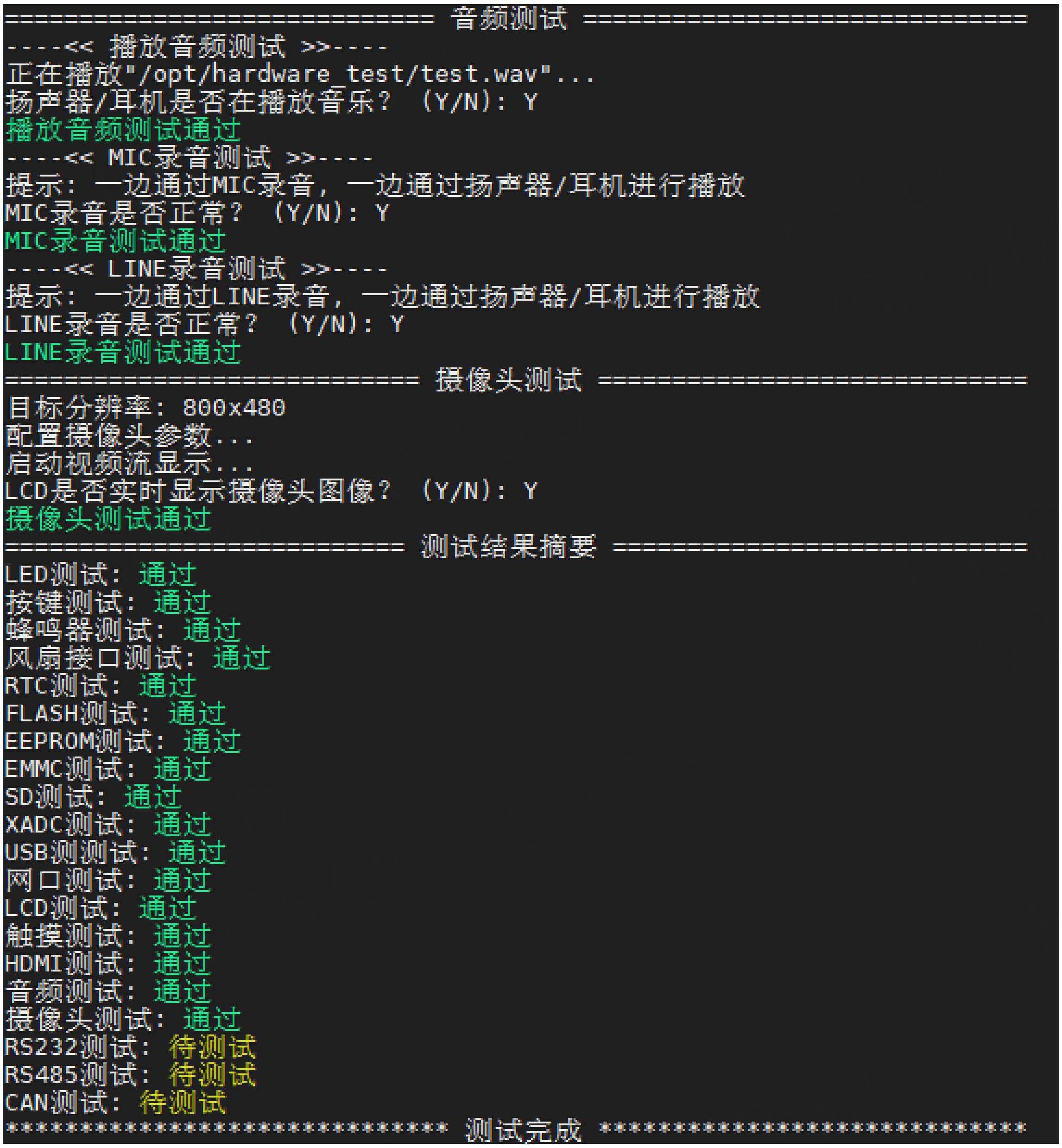
Figure 0-54 Test(1)

```
============================== 蜂鸣器测试 =============================
蜂鸣器是否正常发声？ (Y/N): Y
蜂鸣器测试通过
============================= 风扇接口测试 ============================
开始变速测试(每3秒切换)
当前转速: 255/255
当前转速: 128/255
当前转速: 0/255
风扇转速是否正常？ (Y/N): Y
风扇接口测试通过
=============================== RTC测试 ===============================
设定时间: 2024-10-10 12:00:00
当前时间: 2024-10-10 12:00:01
RTC测试通过
============================== flash测试 ==============================
写flash (mtdblock1)...........................
flash写入成功
读flash (mtdblock1)....
flash读取成功
数据验证...
flash测试通过
============================== eeprom测试 =============================
写EEPROM.....
EEPROM写入成功
读EEPROM....
EEPROM读取成功
数据验证...
EEPROM测试通过
=============================== EMMC测试 ==============================
未检测到分区，检查设备是否已格式化
设备未格式化，正在创建ext4文件系统...
挂载：成功
-------- 顺序写入测试 -------
eMMC写入速度: 19.3 MB/s
-------- 顺序读取测试 -------
eMMC读取速度: 24.0 MB/s
emmc读写测试通过
=============================== SD卡测试 ==============================
检测到已插入SD卡
检测到分区表，使用最后一个分区进行测试
挂载：成功
-------- 顺序写入测试 -------
SD卡写入速度: 16.7 MB/s
-------- 顺序读取测试 -------
SD卡读取速度: 23.7 MB/s
sd卡读写测试通过
============================== XADC测试 ===============================
Temperature: 62.01 °C
vccint: 0.995 V
vccaux: 1.800 V
vccbram: 0.995 V
vccpint: 0.992 V
vccpaux: 1.797 V
vccoddr: 1.466 V
vrefp: 1.239 V
vrefn: -0.008 V
vpvn: 0.011 V
XADC测试通过
=============================== USB测试 ===============================
```

Figure 0-55 Test (2)

```
============================== USB测试 ==============================
----<< USB主机模式测试 >>----
等待将U盘插入到开发板的usb host口(超时:99s)..............
检测到已插入U盘(7.3G)
USB主机模式测试通过
----<< USB从机模式测试 >>----
提示: 先使用USB线连接电脑与开发板的usb slave口
提示: 连接之后, 请观察电脑是否将开发板会识别为U盘
开发板是否被识别为U盘? (Y/N): Y
USB从机模式测试通过
============================= 网口测试 ==============================
在iperf3服务器(电脑)上运行命令: iperf3 -s
请输入iperf3服务器IP地址: 192.168.103.109
----<< PS网口测试 >>----
请把网线插到PS网口中
等待网线连接...........
PS网口的网线已连接
自动获取IP中...........
成功获取IP:192.168.103.158
---- PS网口发送测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PS网口发送速度: 802Mbits/sec
---- PS网口接收测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PS网口接收速度: 790Mbits/sec
PS网口测试通过
----<< PL网口测试 >>----
请把网线插到PL网口中
等待网线连接................
PL网口的网线已连接
自动获取IP中...........
成功获取IP:192.168.103.161
---- PL网口发送测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PL网口发送速度: 575Mbits/sec
---- PL网口接收测试 ----
正在连接服务器192.168.103.109...
已连接192.168.103.109服务器, 开始测试...
PL网口接收速度: 549Mbits/sec
PL网口测试通过
=========================== LCD显示测试 ============================
提示: 5秒后关闭显示, 请观察lcd屏
LCD是否正常显示? (Y/N): Y
LCD测试通过
============================ 触摸屏测试 ============================
---- 多点触摸测试模式 ----
提示: 点击"Draw", 然后多根手指按下并滑动
多点触摸是否正常? (Y/N): Y
触摸屏测试通过
请点击"Quit"退出触摸测试
=========================== HDMI显示测试 ===========================
提示: 5秒后关闭显示, 请观察HDMI屏
HDMI是否正常显示? (Y/N): Y
HDMI测试通过
============================= 音频测试 ==============================
```

Figure 0-56 Test (3)



```
============================== 音频测试 ===============================
----<< 播放音频测试 >>----
正在播放"/opt/hardware_test/test.wav"...
扬声器/耳机是否在播放音乐？ (Y/N): Y
播放音频测试通过
----<< MIC录音测试 >>----
提示: 一边通过MIC录音，一边通过扬声器/耳机进行播放
MIC录音是否正常？ (Y/N): Y
MIC录音测试通过
----<< LINE录音测试 >>----
提示: 一边通过LINE录音，一边通过扬声器/耳机进行播放
LINE录音是否正常？ (Y/N): Y
LINE录音测试通过
============================= 摄像头测试 ==============================
目标分辨率: 800x480
配置摄像头参数...
启动视频流显示...
LCD是否实时显示摄像头图像？ (Y/N): Y
摄像头测试通过
============================ 测试结果摘要 ============================
LED测试: 通过
按键测试: 通过
蜂鸣器测试: 通过
风扇接口测试: 通过
RTC测试: 通过
FLASH测试: 通过
EEPROM测试: 通过
EMMC测试: 通过
SD测试: 通过
XADC测试: 通过
USB测试: 通过
网口测试: 通过
LCD测试: 通过
触摸测试: 通过
HDMI测试: 通过
音频测试: 通过
摄像头测试: 通过
RS232测试: 待测试
RS485测试: 待测试
CAN测试: 待测试
****************************** 测试完成 ******************************
```

Figure 0-57 Test (4)

Among them, the RS232, RS485 and CAN tests have not been tested yet, because these three tests need to communicate with other devices and are inconvenient to test. These three tests are explained in the "Individual Testing" section.